OMDAYAL GROUP OF INSTITUTIONS

COMPUTER SCIENCE & ENGINEERING

LLM Integrated OCR for Medical Document Analysis and Profile Generation

OCR-first medical document understanding with an intelligent LLM fallback for difficult or uncertain text.

OCR-FIRST PIPELINE | LLM FALLBACK | MEDICAL PROFILE | RECORD LINKING

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TECHNOLOGY STACK

- Python
- OpenCV
- OCR / HTR
- Vision LLM
- NLP / Transformers
- FastAPI
- React.js

How we prepare pixels before OCR

Gaussian adaptive thresholding helps separate text from uneven medical-document backgrounds before text detection.

PREPROCESSING PIPELINE

1. **Grayscale**
   Convert colour image to intensity values
2. **Denoise**
   Reduce scanner / background noise
3. **Gaussian Adaptive Threshold**
   Compute a local threshold for each pixel
4. **Deskew / Normalize**
   Correct rotation and improve text alignment

Goal: make text pixels easier to detect.

GAUSSIAN ADAPTIVE THRESHOLDING

Instead of one global threshold, each pixel gets a threshold from its local neighbourhood.

$$T(x,y) = G(x,y) - C$$

$$G(x,y) = \Sigma\, w_i \cdot I_i$$

$$B(x,y) = 1 \text{ if } I(x,y) \geq T(x,y)$$
$$B(x,y) = 0 \text{ otherwise}$$

w_i = Gaussian weights
I_i = neighbouring pixel intensities
C = small constant

Local threshold → adapts to uneven lighting.

MATHEMATICAL EXAMPLE

3 × 3 local pixel windows

100	110	120
110	150	130
90	100	110

Gaussian kernel

$$\begin{bmatrix} 1 & 2 & 1 \\ 2 & 4 & 2 \\ 1 & 2 & 1 \end{bmatrix} / 16$$

$G = 1920 / 16 = 120$

Let $C = 10$ → $T = 120 - 10 = 110$

Center pixel = 150
$150 \geq 110$ → foreground (1)

From visual patterns to a text sequence

The recognition stage learns a mapping between image features and possible character / token sequences.

RECOGNITION — CONCEPTUAL VIEW

Visual input

A m o x i c i l l i n

Feature extraction

edges • shapes • strokes • spatial patterns

Sequence probabilities

P(text | image)

The model estimates which text sequence best explains the visual evidence.

THREE DIFFERENT TASKS

WHERE?

Text Detection

Find the regions / lines that contain text.

WHAT?

Recognition

Map visual patterns to likely characters / tokens.

WHICH SEQUENCE?

Decoding

Select the most plausible sequence from model probabilities.

Medical meaning is NOT produced here.

Meaning comes next → LLM / NLP → structured medical profile

OCR → LLM → Medical Profile Generation

OCR is the fast first-pass reader, the Vision LLM acts as a fallback and semantic reasoning layer.

STEP 1 • DOCUMENT

CITY CARE CLINIC

Patient: Rahul Kumar

Date: 12 / 01 / 2026

Rx: Amoxicillin 500 mg

1 tablet • 2 times/day

Follow-up: 7 days

STEP 2 • OCR

Patient: Rahul Kumar

Date: 12 / 01 / 2026

Rx: Amoxicillin 500 mg

1 tablet • 2 times/day

OCR confidence

$C_o^{cr} = 0.96$

High confidence → continue

STEP 3 • LLM

If OCR is weak:

image → Vision LLM

The LLM can inspect the visual context and produce a structured interpretation.

Fallback path when OCR fails

STEP 4 • PROFILE

Patient	Rahul Kumar
Medicine	Amoxicillin
Dose	500 mg
Schedule	2 / day
Follow-up	7 days

Structured medical profile

IMAGE → PREPROCESS + GAUSSIAN ADAPTIVE THRESHOLD → OCR → CONFIDENCE GATE → [LLM FALLBACK if needed] → MEDICAL PROFILE → RECORD LINKING → TIMELINE